



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEK D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN *(INFORMATION SHEET)*

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-4: 2015 CORE ABILITIES	
TAHAP / LEVEL	4 (FOUR)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA05 STRATEGIC THINKING SKILL	
NO DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	05.01 ADVOCATE EFFECTIVE THINKING 05.02 COMPLY WITH EFFECTIVE COMPETENCY ANALYSIS 05.03 APPLY STRESS MANAGEMENT TECHNIQUES 05.04 APPLY BUDGETARY CONTROL PROCEDURES 05.05 APPLY PROBLEM SOLVING TECHNIQUES	
NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 1 Drpd / of 12

TAJUK/TITLE: EFFECTIVE COMPETENCY ANALYSIS

TUJUAN/PURPOSE:

This Information Sheet (IS) is intended to describe the knowledge, skills and attitude requires to advocating effective thinking in strategic thinking skills that include knowledge of thinking principles in relation to issues, ideas and its application to problem solution. The person who is competent in this CA shall be able to comply with effective analysis in thinking strategies.

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 2 Drpd. / of: 12
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PENERANGAN/INFORMATION:

1. Competency

1.1 Competency" is a cluster of related knowledge, skills, and attitudes that affects a major part of one's job (a role or responsibility), that correlates with performance on the job, that can be measured against well-accepted standards, and that can be improved via training and development. For example - you might have good interpersonal skills (skills), but will not be competent to join a company as Project Manager unless you possess adequate education / experience (knowledge) and the right temperament / behaviour (attitude).

1.2 Competency-based interviews (also called structured interviews) are interviews where each question is designed to test one or more specific skills. The answer is then matched against pre-decided criteria and marked accordingly. For example, the interviewers may want to test the candidate's ability to deal with stress by asking first how the candidate generally handles stress and then asking the candidate to provide an example of a situation where he worked under pressure.

1.3 Normal interviews (also called unstructured interviews) are essentially a conversation where the interviewers ask a few questions that are relevant to what they are looking for but without any specific aim in mind other than getting an overall impression of you as an individual. Questions are fairly random and can sometimes be quite open. For example, a question such as "What can you offer our company?" is meant to gather general information about you but does not test any specific skill or competency. In an unstructured interview, the candidate is judged on the general impression that he/she leaves; the process is therefore likely to be more subjective.

1.4 Competency-based interviews (also called structured or behavioural interviews) are more systematic, with each question targeting a specific skill or competency. Candidates are asked questions relating to their behaviour in specific circumstances, which they then need to back up with concrete examples. The interviewers will then dig further into the examples by asking for specific explanations about the candidate's behaviour or skills.

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 3 Drpd. / of: 12
--------------------------	--------------------------------	--

1.5 Many interviews now are competency-based which means that the interviewer will be looking for you to answer questions about your abilities and experience in the context of actual events. It is worth preparing in advance at least two examples for each competency as it is not unheard of for the recruiter to ask for a second example, particularly if you have already quoted one on an application form.

1.6 The list of skills and competencies that can be tested varies depending on the post that you are applying for. For example, for a Personal Assistant post, skills and competencies would include communication skills; ability to organise and prioritise; and ability to work under pressure. For a senior manager, skills and competencies may include an ability to influence and negotiate; an ability to cope with stress and pressure; an ability to lead; and the capacity to take calculated risks.

2. Ideas

1.1 Ideas are usually construed as mental representational images of some object. Ideas can also be abstract concepts that do not present as mental images. Many philosophers have considered ideas to be a fundamental ontological category of being. The capacity to create and understand the meaning of ideas is considered to be an essential and defining feature of human beings. In a popular sense, an idea arises in a reflexive, spontaneous manner, even without thinking or serious reflection, for example, when we talk about the *idea* of a person or a place.

2.2 Generating Idea

The process of creating, developing, and communicating ideas which are abstract, concrete, or visual is known as idea generation. The process includes the process of constructing through the idea, innovating the concept, developing the process, and bringing the concept to reality. An idea is usually generated with intent, but can also be created unintentionally. Ideas often form:

- i) During brainstorming sessions or through discussions.
- ii) As an impression or notion that tries to portray the overarching scope or outline.
- iii) As an opinion or belief.

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 4 Drpd. / of: 12
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Figure 1.2: Overloading of Ideas

2.3 Information

Information is facts provided or learned about something or someone. Information is conveyed or represented by a particular arrangement or sequence of things and has meaning in some context for its receiver.

2.4 Language and Tone.

Tone is the use of pitch in language to distinguish lexical or grammatical meaning – that is, to distinguish or to inflect words. All verbal languages use pitch to express emotional and other paralinguistic information and to convey emphasis, contrast, and other such features in what is called intonation, but not all languages use tones to distinguish words or their inflections, analogously to consonants and vowels.

Languages that do have this feature are called **tonal languages**; the distinctive tone patterns of such a language are sometimes called **tonemes**.

2.5 What does it take to communicate with another person?

We communicate with much more than our words. In face-to-face communication, words are only part of the message. The balance of the message, and in fact, the largest part of the message send to others is made up of non-verbal information. It is composed of two communication elements of body language and tone of voice that are:

- i) Face to face communication

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 5 Drpd. / of: 12
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- ii) Tone of voice - the tone of voice use in communication is responsible for about 35-40 percent of the message we are sending. Tone involves the volume you use, the level and type of emotion that you communicate and the emphasis that you place on the words that you choose.

2.5 Ideas and Information Sources

Sources of ideas and information are useful in project activities such as concept generation, detail design, brainstorming, evaluation matrices, quality function deployment and morphological methods. These sources include:

i) Books:

- a) Handbooks: Handbooks on data, procedures and technical information covering various topics such as materials, finishing and coating, machining, and design for manufacturability.
- b) Textbooks: Books are frequently a good source of information. Although technical information may not be the most up to date by the time a book is published, even some very old texts can sometimes provide or at least stimulate some interesting ideas. Some old ideas can find a new life through novel materials, processing or design methods.
- c) Industry manuals: Many engineering associations produce manuals that provide general (e.g., design procedures) and specific (e.g., motors, bearings) information about standard products, parts and procedures. Nuts and bolts, for example, are described in the SAE Handbook.
- d) Periodicals and Technical Reports:
 - Trade Magazines: There are a number of trade magazines that cover general design issues or are targeted at a specialized technical field. These magazines are often a source of solutions for current problems. Besides the articles in these magazines, the advertisements can provide a fruitful source of ideas. As it is difficult to pinpoint specific information in trade magazines, it

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 6 Drpd. / of: 12
--------------------------	--------------------------------	--

is a good practice to make a habit of regularly reading or at least scanning these publications so information can be located when required.

- Research Journals: Research journals directly related to the specific area of the problem to be solved may provide modelling and analysis of specific problems as well as more general information that can lead to a design solution.

e) Databases: Databases provide a mechanism by which articles from hundreds of technical journals spanning numerous years can be search for a specific subject. The usability of these databases has been greatly enhanced in recent years by computerization. Examples of these databases include:

- The Engineering Index and The National Technical Information Service. These databases can be searched by general categories or specific key words can be used for more targeted searches. Directories
- Thomas Register: The Thomas Register of American Manufacturers is a set of volumes that provide information about manufacturers of a multitude of products and services. It can typically be found in the library but is now conveniently available on the Internet at www.thomasregister.com.
- Fraser's Canadian Trade Directory: Similar to the Thomas Register, Fraser's Canadian Trade Directory provides information about Canadian providers of products and services. It is available in both hard copy and CD-ROM formats, and can also be accessed through the Internet at www.frasers.com where searches can be conducted within the categories of company, product/service or brand name.
- Yellow Pages: The yellow pages are another source for suppliers and manufacturers. If availability from local suppliers is insufficient, yellow pages for other cities can often be found at the library or on the Internet.
- Catalogues: There are hundreds of catalogues of parts, assemblies and materials available through vendors. These catalogues are often available through workshops and resource centres, or can be ordered by mailing away request cards often included in trade publications. Other Literature Sources
- World Wide Web: Searching the Internet can lead to websites belonging to vendors, manufacturers, consultants, design companies, professional organizations and educational centres, to name a few.

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 7 Drpd. / of: 12
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- Brochures: Marketing_publications available from competitors or for related products can sometimes provide valuable information.

ii) People as Idea and Information Sources: An extremely valuable source of ideas that often gets overlooked is people. It is truly amazing the viable ideas that can come from not only subject experts, but also the most unlikely sources including the person sitting next to you on the plane or a small child. Designers should never be afraid to ask others for ideas. Even if someone does not have any solutions to offer, they may know where to locate information or a person who can provide information. This is called networking and can be one of the most powerful tools in business today.

- Colleagues: Colleagues are usually a very approachable source of ideas. They may be people within the same company, contacts within other organizations or former classmates.
 - Consultants: Consultants are generally people who are experts in a certain area. Although mostly paid for there expertise, many willingly will answer questions and provide information as long as it is not in conflict with their commitments and doesn't require a great deal of time. Consultants can generally be located through word-of-mouth, professional directories, yellow pages, educational institutions or the Internet.
 - Vendors: Although vendors typically are interested in selling their product, many value the development of relationships and will therefore often be a willing source of ideas and even samples. Vendors are an excellent resource as they often are current in technology and have the opportunity to visit a variety of companies where they can view different ways of doing things.
 - Customers: Customers are an excellent source of ideas. Often they have conducted their own research before committing to working with a team of designers on a specific project. They may have seen similar products in use and have ideas on different concepts that could be employed.
 - Lead users: These people are those that had a need for a product long before the general population. Lead users have often created adaptations or invented their own solutions to satisfy their requirements until a new product comes along.
- Friends/family: Many designers find ideas for projects when casually talking to friends and family. Sometimes people who have little knowledge of

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 8 Drpd. / of: 12
--------------------------	--------------------------------	--

the technology or the specific problem to be solved may look at things from a different perspective and propose creative solutions.

iii) Organizations as Idea and Information Sources

- a) Government: There are numerous sources within all levels of government have standards and guidelines based on extensive research.
- b) Professional Organizations: Groups such as Association of Professional Engineers often have publications and codes available. This organization typically can identify and provide contact information for experts on a given subject.

iv) Other Idea and Information Sources

- a) Nature: Many innovative technical solutions are based on principles found in nature. Birds have provided concepts for winged flight as fish have for submarines. Honeycombs have provided examples of lightweight but strong structures and the arch of the human foot demonstrates the effectiveness of a keystone bridge structure for supporting heavy loads. Velcro™, an extensively used fastening device, was based on the gripping capabilities of a burr.
- b) Patents: Patents are an excellent source of technical ideas. Although the use of protected products may be prohibited or require the payment of a royalty, expired and foreign patents without global coverage can be used. Reviewing restricted patents may spawn innovative new ideas that do not fall under the restriction of a patent.
- c) Analogies: The creative technique of using analogies involves examining the similarities between the current design problem and another similar problem.

2.6 Analytical skills?

Analytical skills are the ability to visualize, gather information, articulate, analyse, solve complex problems, and make decisions. Such skills include demonstration of the ability to apply logical thinking to gathering and analysing information, designing and testing solutions to problems, and formulating plans. This skills are essential in the workplace to

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 9 Drpd. / of: 12
--------------------------	--------------------------------	--

ensure necessary problem solving occurs to keep productivity and other areas of the workforce functioning smoothly.

In all instances analytical skills require you to dissect a problem and then find a solution for that problem. There are many skills required within a company and its workforce, but analytical skills are essential in every organization.

i) Testing Analytical Skills

If you were to test for analytical skills you might be asked to take a series of events and put them in proper sequence, look for advertisement inconsistencies, or read an essay with a critical eye. There are generally standardized tests used as a guideline. There are many examples of analytical skills that can be applied in a work environment. A few examples include:

- a) Analysing abilities: The ability to take a large volumes of data and then analyse trends and produce a result
- b) Dealing with problems: Give examples of problem solving at work
- c) Programming: Write a systems program with accurate results output
- d) Reporting: A written report on the effectiveness of a particular event such as a political campaign.
- e) Handling assignments effectively: Discovering a more efficient and productive way to complete a particular job task.
- f) Process: Creating a set of steps to implement a process that could have a yes or no outcome.
- g) Problem handler: Identifying a problem and then creating a repair to avoid it becoming a major problem.
- h) Collect information: Analyze the data from the information collected, and come up with a solution to a problem.

2.7 Effective Analysis

Although many ideas result from brainstorming, storyboarding and other creative processes, the majority either stem directly from or are based on existing ideas from various sources. For example, an analysis of competitive, legacy or similar products, or individual functions or components of these products, may lead to design solutions.

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 10 Drpd. / of: 12
--------------------------	--------------------------------	---

There are a number of approaches for organizing, assessing manipulating ideas as describe in Pahl and Beitz such as systematic combination, and combining with the help of mathematical methods and Hyman - morphological boxes and charts. The use of checklists is another technique for creative thinking and manipulating ideas. Checklist A provides an example of an idea-generating checklist suggests some basic guidelines to follow when attempting to locate information:

Nos.	Basic Guidelines
1.	Take the time to determine clearly the information that is required.
2.	It always takes longer than anticipated to find information. Be generous with time allotments.
3.	Determine what is already known such as individuals or companies working in the field of interest. Use relevant dates to help with the search (e.g., information on lasers will likely be found in publications from 1960 and newer).
4.	Consult the easiest and most accessible tools first.
5.	Be as specific as possible. However, if a specific search proves futile, a broader search may have to be employed.
6.	Do not stop with a single reference. Search also for newer and alternate sources.
7.	If a search is going nowhere, re-evaluate the search criteria and/ or sources. • It is better to ask for help than waste time. Other hints that should be considered when searching for ideas and information include:
8.	Consider the source. If information is obtained from a periodical at a grocery checkout counter or a personal website, it is advisable to seek other sources to confirm or refute the validity of the information. It is even advisable to consult multiple sources to confirm information obtained from reputable sources.
9	Weigh the value of the information against
10.	Peruse cited references within documents for other potential sources of information.

2.8 Effective analysis strategies.

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 11 Drpd. / of: 12
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Effective Analysis (EA) involves deciphering the consequences of those break downs by making sure that all failures can be detected, determining how frequently a failure might occur and identifying which potential failures should be prioritized.

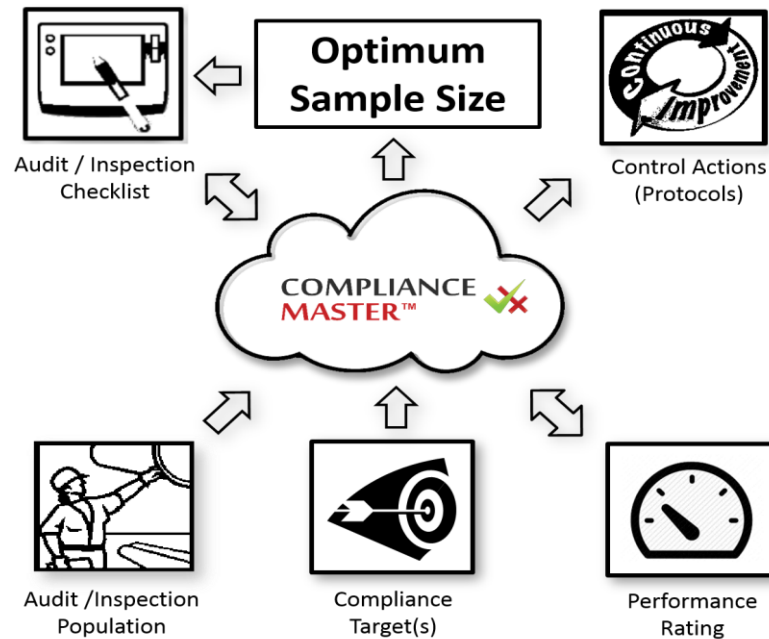


Figure 1.8: Compliance to Effective Analysis

SOALAN/QUESTION:

1. Explain the different between ideas and information.
2. What do you understand with analytical skills?
3. List 10 basic guidelines that you might consider in your idea generating checklist of effective analysis.
4. With the aids of a diagram, describe the strategies you might use to comply with effective analysis.
5. Explain briefly about competencies?

NO. KOD ICODE NO.	Z-009-4:2015-M05/P(2/5)	Mukasurat / Page : 12 Drpd. / of: 12
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